

Wellbotics
Queen Game
Sample Curriculum

Overview: How Does Your Garden Grow

Target Audience:

Learners aged 5 - 8 years (K-2).

Wellbotics programs are designed to be developmentally appropriate, ensuring targeted instruction for this age group.

Overarching Goal:

The "Queen" game aims to foster academic growth, social and emotional learning (SEL), and digital fluency by integrating STEM activities, coding, and robotics with SEL principles. Wellbotics utilizes the analogous relationship between robots and humans to facilitate mindfulness and help children apply what they learn from the robot realm to their own lives, leveraging coding and robotics to represent human thought and behavior. Educational games are effective tools for learning, increasing motivation, emotional involvement, interaction, and enjoyment.

I. Core Learning Goals & Standards Alignment

The curriculum focuses on two key areas: programming basics (sequencing and debugging) and Social and Emotional Learning (SEL), including self-awareness, persistence, and mindfulness practice.

A. STEM & Computer Science (CS) Education

The game's activities are directly supported by national and state computer science and engineering standards for K-2 grade levels, including the Computer Science Teachers Association (CSTA) K-12 Standards Level 1A, New York State Computer Science and Digital Fluency Learning Standards K-2, and principles of Next Generation Science Standards (NGSS K-2 ETS) related to engineering design.

1. Programming Basics: Sequencing & Algorithms

- Objective: Students will understand that computers follow step-by-step instructions to solve problems. They will learn to create and follow simple algorithms.
- Standard Alignment: CSTA K-12 CS Standards Level 1A: "Create programs to accomplish tasks as a means of problem solving or artistic expression" (e.g., sequencing directional commands). NYS K-2.AP.1: "Develop and use algorithms to complete a task".
- K-2 Progression:
 - Kindergarten: Students use direction cards with visual cues and focus on sequencing 1-2 steps.
 - 1st Grade: Students introduce basic command blocks or cards and are encouraged to chain 2-3 steps logically.
 - 2nd Grade: Students are allowed light exploration of loops or simple conditional instructions (e.g., 'if blocked, turn left').
- Game Application: Players program their Bee-Bots to move through the garden, collecting pollen and returning to the hive, which directly involves creating and following step-by-step instructions.

2. Debugging

- Objective: Students will identify and fix errors in their programs when unexpected outcomes occur.
- Standard Alignment: NYC CS4All Blueprint K-2: "Decomposition: Break down complex problems into smaller, manageable parts."
- K-2 Progression:
 - K-1: Offer physical manipulatives and scaffold with "if-then" statements. Encourage prediction: "What will happen if...?".
 - 2nd Grade: Students are allowed to debug a sequence or revise strategy when their initial program doesn't work.
- Game Application: Obstacle/Movement Limitation Cards introduce challenges that require players to adapt their strategies and "reprogram" their Bee-Bots.

3. Computational Thinking & Problem-Solving

- Objective: Students will break down complex tasks into smaller, manageable steps and devise solutions.
- Standard Alignment: NYS K-2.CT.1: "Students use step-by-step instructions to demonstrate the flow of actions, aligning decisions with computational steps".
- Game Application: Players learn to break down complex tasks, like navigating the garden, into smaller parts. The game encourages students to "identify problems" and learn "problem-solving strategies" within the coding context.

4. Physical Computing

- Objective: Students will apply their coding knowledge in a tangible way by controlling physical robots.
- Game Application: The use of Bee-Bots allows students to see their algorithms come to life in the physical game space, moving through the "garden" to collect "pollen".

B. Social-Emotional Learning (SEL)

The "Queen" game deeply integrates SEL principles, drawing on widely accepted frameworks like CASEL's 5 Core Competencies (Self-Awareness, Self-Management, Social Awareness, Relationship Skills, and Responsible Decision-Making) and New York State SEL Benchmarks for K-2. Wellbotics emphasizes that coding and robotics provide a powerful way to understand oneself and the world, leveraging the analogous relationship between robots and humans to foster emotional intelligence and problem-solving skills.

1. Self-Awareness & Self-Management (Persistence & Mindfulness)

- Objective: Students will recognize their own emotions and how they influence their actions, and develop strategies for emotional regulation and persistence in the face of challenges.
- Standard Alignment: NYS SEL Benchmark 1A: "Students can recognize and name basic emotions". NYS SEL Benchmark 1B: "Students recognize how their feelings affect actions".

- K-2 Progression:
 - Kindergarten: Model and name emotions using the bee roles (worker, scout, queen).
 - 1st Grade: Ask students to identify when the hive is 'stressed' or 'healthy' and reflect on how their choices affect it.
 - 2nd Grade: Encourage emotional journaling or role-play (e.g., 'How would a lost scout feel?').
- Game Application:
 - "Dance Cards" prompt players to connect movement to emotional states, such as showing how they feel when the hive is "strong". Wellbotics explicitly aims to help children express feelings through robotics.
 - The "Frustration Station!" card directly addresses emotional regulation and persistence by encouraging players to "Take a deep breath" and "re-program".
 - Mindfulness practice is introduced through "Mindful Moment Dance!" cards, guiding students to move slowly and notice their bodies, reflecting on what "calm" feels like.

2. Social Awareness & Relationship Skills (Collaboration & Communication)

- Objective: Students will understand and empathize with others' perspectives, and develop skills for effective teamwork and communication.
- Standard Alignment: CASEL's framework includes social awareness and relationship skills. NYS K-2 SEL Goal 2.
- Game Application:
 - The game is designed for "ongoing collaborative play" to maintain hive health, which directly fosters teamwork, communication, sharing, and negotiation.
 - "Dance Cards" like "Share Your News Dance!" and "Teamwork Wiggle!" explicitly promote communication and cooperation, encouraging students to act out emotions and create new moves that show collaboration.
 - Students learn to "recognize different responses to various situations" as they discuss how sound affects emotions, promoting social awareness.

3. Responsible Decision-Making

- Objective: Students will identify problems and make choices that contribute to the well-being of their community.
- Standard Alignment: CASEL's framework includes responsible decision-making. NYS K-2 SEL Goal 3.
- Game Application: By facing obstacles and problem-solving within the game, students develop the ability to identify problems and make choices that contribute to a positive and inclusive community (the hive). Emphasis on collaborative problem-solving, with prompts like "Problem! How can you solve this with your team?", reinforces these decision-making skills.

II. Game Components & Gameplay Flow

- Game Board: Represents a garden divided into tiles. Tiles can potentially be movable for customization.
- Bee-Bot: Each player controls a Bee-Bot, programming it to move through the garden.
- Queen Bee (Facilitator): The facilitator guides gameplay, provides instructions, and visually demonstrates hive health (e.g., costume, gestures, visual indicators).
- Hive (Starting Space): The safe home base where Bee-Bots return with collected pollen. Its placement can adjust game difficulty.
- Objective: To maintain and improve Hive Health through ongoing collaborative play, by effectively collecting pollen and performing dances to communicate. There is currently no defined winning state; instead, the focus is on sustained hive health.

Card Types: Cards are a fundamental component of tabletop games and are designed with visual dominance, color-coding, and tangible elements to optimize engagement and clarity for young learners.

1. Collect Pollen Cards:

- Visuals: Bright, engaging illustrations of different pollen types or happy flowers.
- Action/Prompt: Simple instructions like "Collect Yellow Pollen!" or "Found Sparkling Pollen!". Players attach pollen directly to their Bee-Bot, which is then returned to the hive to improve health.
- SEL Integration: Narration like "The hive feels happy when you bring pollen!" or "Your pollen helps the hive feel strong!" implicitly reinforces collaboration and individual contributions to collective well-being, fostering social awareness and positive relationship skills.

2. Obstacle/Movement Limitation Cards:

- Visuals: Clear icons depicting garden obstacles (e.g., fallen branch, gust of wind, puddle, tangled vines).
- Action/Prompt: Specific instructions like "Windy Path! Move back 1 space," "Tangled Vines! Use 1 extra programming card this turn," or "Lost Pollen! Oh no! You drop 1 pollen. Can you re-plan?". These challenges require players to adapt strategies or reprogram.
- SEL Integration: Directly address perseverance and problem-solving with prompts like "Bumpy Road! It's tough, but don't give up!" and "Frustration Station! Take a deep breath. Can you re-program?". Collaboration is encouraged: "Problem! How can you solve this with your team?".

3. Dance Cards:

- Visuals: Dynamic illustrations of bees/Bee-Bots dancing, and emoticons reflecting emotions (e.g., happy, thinking, curious).
- Action/Prompt: Commands such as "Perform the 'Happy Hive Dance!'" or "Show the 'Curiosity Crawl!'". These involve collaborative physical movements and develop psychomotor skills.
- SEL Integration: Connect movement to emotional states: "Dance to show how you feel when the hive is strong!". Promote self-awareness and communication: "Share Your News Dance! Act out how you feel about collecting pollen. Can your friends guess?". Reinforce collaboration and relationship skills: "Teamwork Wiggle! Work together to create a new move that shows cooperation". Introduce mindfulness: "Mindful Moment Dance! Move slowly and notice your body. What does calm feel like?". The facilitator plays a crucial role in debriefing, linking robot actions to human behaviors and emotions.

Sequence of Play:

1. Queen Bee Gives Instructions: The Queen provides players with a destination tile and a maximum number of programming cards allowed.
2. Players Program Bee-Bots: Players use their programming cards to plan their Bee-Bot's route.
3. Bee-Bots Execute Program: Players run their programmed Bee-Bots on the game board.

4. Card Interactions: Players land on tiles, draw and interact with Pollen, Obstacle, or Dance cards.
5. Dance (Optional/Triggered Event): Players perform dances at the hive to enhance hive health and reinforce SEL through facilitated reflection and movement.
6. Assess Hive Health: The Queen visually demonstrates current hive health, indicating the effectiveness of pollen collection and communication.

This integrated approach aims to provide a hands-on, engaging, and developmentally appropriate learning experience that fosters essential STEM and SEL skills in young learners.